

Metrocar App Analytics

Metrocar is a ride-hailing application that connects drivers and passengers.

Project Goal is to analyze key performance metrics of the application, including:

- building user and ride funnels;
 - analyzing the relationship between revenue and time of day or day of the week;
 - identifying weak points and determining where the highest user drop-off occurs;
 - analyzing which platform is the most effective and generates the highest revenue.
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Data

The following tables were analyzed in this project:

- *app_downloads* — data on app downloads;
 - *signups* — information about new user registrations;
 - *ride_requests* — ride request data;
 - *transactions* — financial transaction data;
 - *reviews* — user feedback and ratings after rides;
 - *funnel_analysis* — a pre-built table combining multiple datasets, containing counts of downloads, signups, rides, payments, and reviews.
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Key Metrics

- Total revenue — \$4.25M
- Number of users — 17,623
- Number of successful rides — 223,652

- Average driver rating — *3.06*
- Average driver wait time — *8.5 min*
- Average ride request acceptance time — *6.9 min*

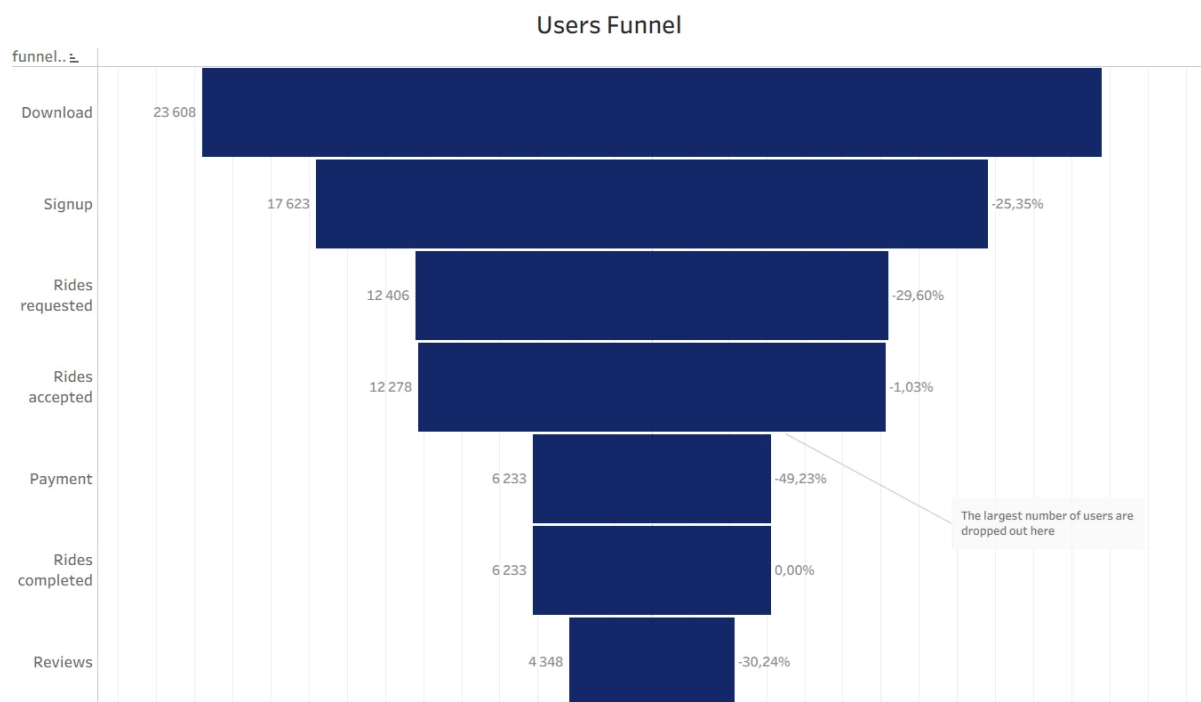
Two filters were created:

- platform
- age range

User Funnel

Steps:

Download → Signup → Ride Requested → Ride Accepted → Payment → Ride Completed → Review



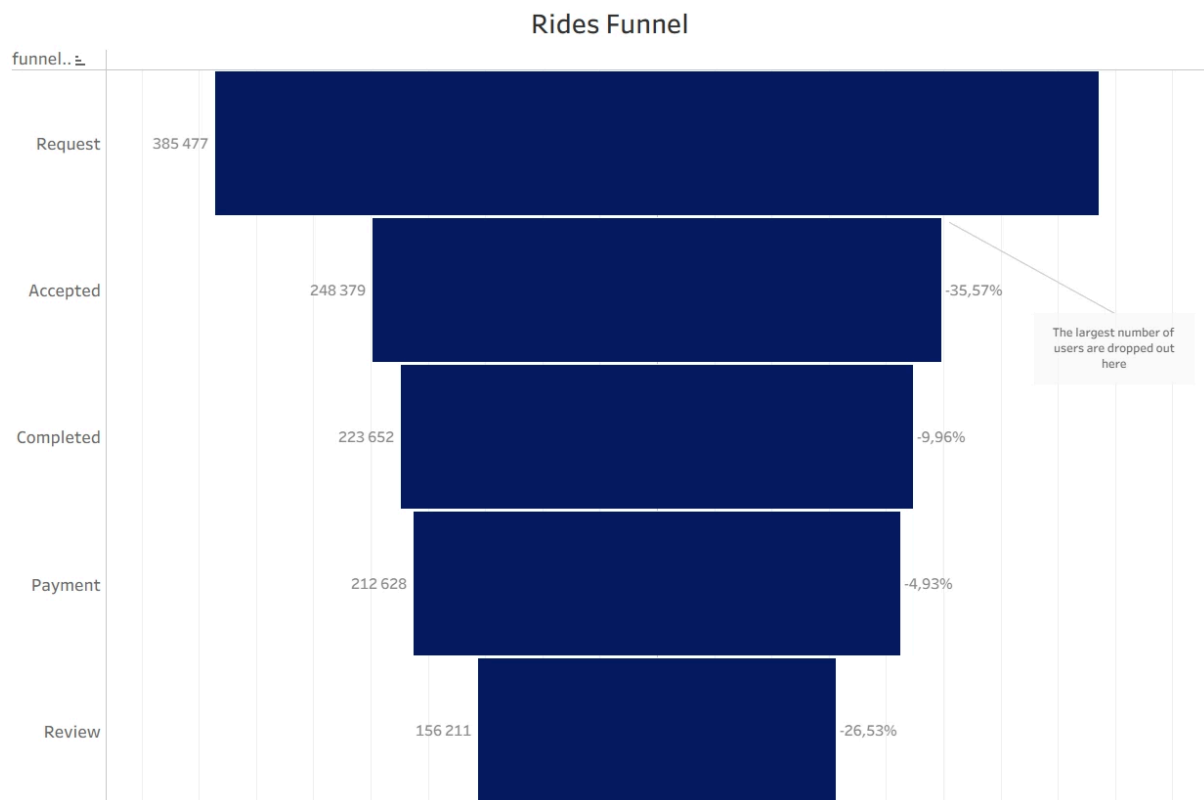
Key Insights:

- the highest user drop-off occurs between *Ride Accepted* → *Payment* (-49.23%);
- a significant drop-off is also observed at *Ride Completed* → *Review* (-30.24%).

Ride Funnel

Steps:

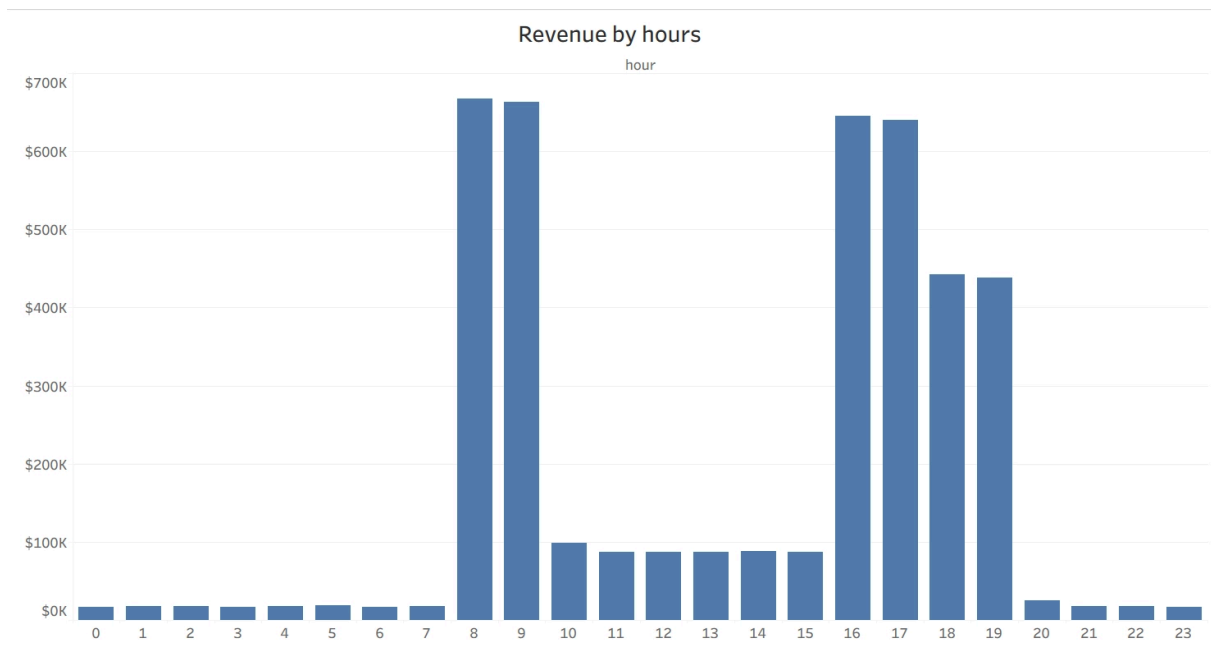
Request → Accepted → Completed → Payment → Review



Key Insights:

- the largest drop-off occurs at the first stage *Request* → *Accepted* (-35.57%);
- a significant drop-off is also seen at *Payment* → *Review* (-26.53%).

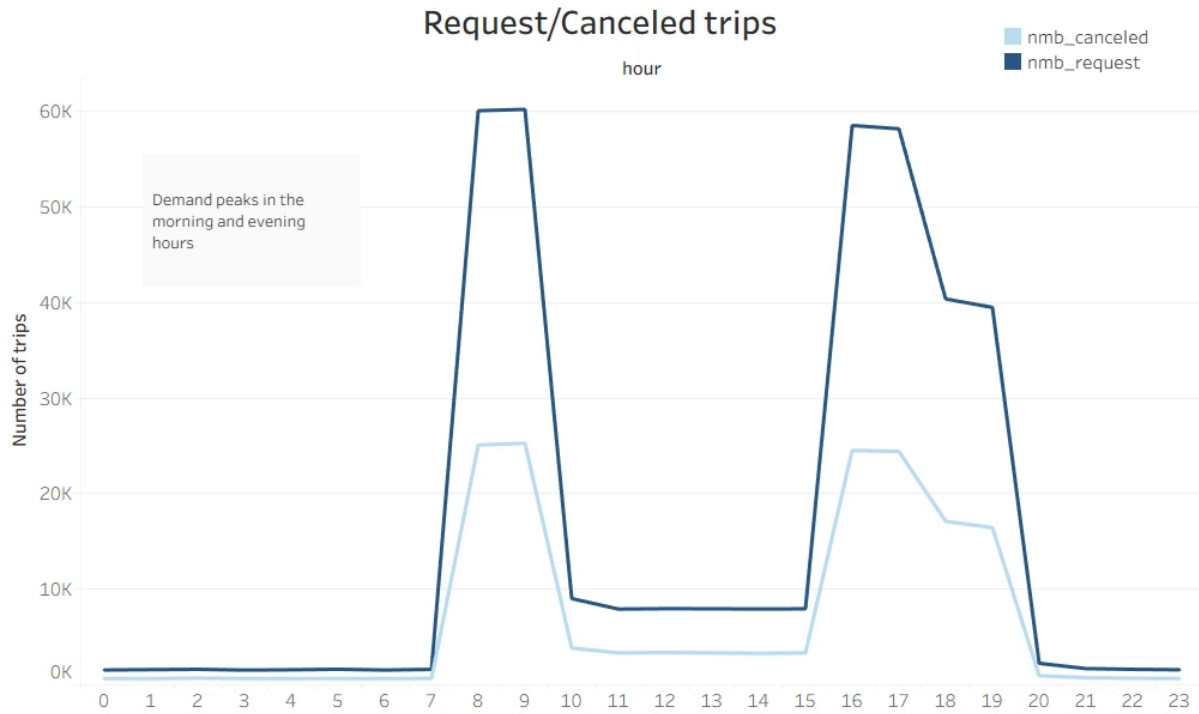
Revenue by Time of Day



Key Insights:

- peak revenue occurs during morning and evening hours, specifically 8:00–9:00 and 16:00–19:00;
- there is a noticeable decline between 10:00 and 15:00;
- nighttime hours generate minimal revenue.

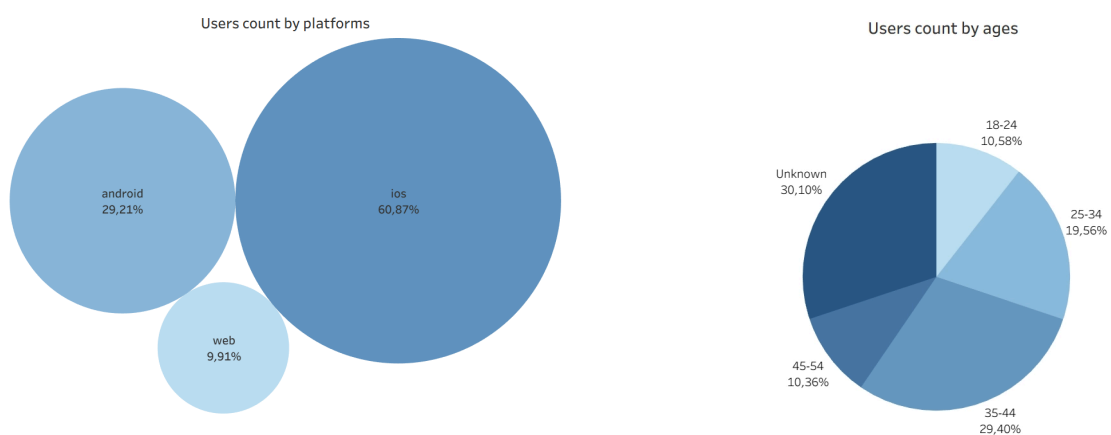
Ride Requests vs Cancellations by Hour



Key Insights:

- the highest number of ride requests occurs during morning and evening peak hours (8:00–9:00 and 16:00–19:00);
- cancellations follow the same pattern, peaking during these hours as well.

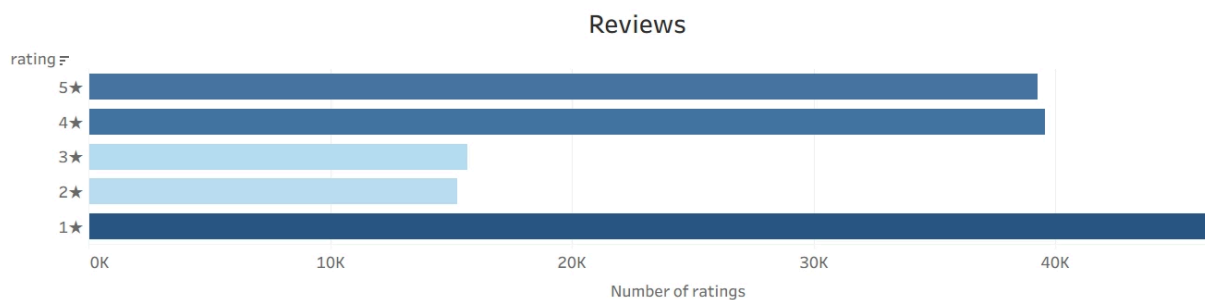
User Distribution by Age and Platform



Key Insights:

- the majority of users are on the **iOS** platform (~61%), while **web** has the smallest share (~10%);
- about one-third of users have **unknown age** (~30%);
- the largest age group is **35–44 years old** (~30%).

Ratings and Reviews



Key Insights:

- users most frequently give **1-star ratings**;
- the second most common ratings are **4 and 5 stars**.

Key Insights Summary

1. The largest user drop-off occurs between ride request and payment — 49%.
2. A significant drop occurs between request and acceptance — 36%.
3. The highest revenue is generated during morning and evening peak hours, which aligns with commuting patterns.
4. A large number of ride requests occur during peak hours, but cancellations are also high, suggesting a possible driver shortage and lost revenue.
5. One-third of users did not provide their age, but the largest known group is 35–44 years old.
6. iOS is the dominant platform, generating 61% of revenue.
7. Users tend to leave ratings only when they feel strong positive or negative emotions.

8. The average rating is 3 stars, likely influenced by long wait and acceptance times (8.5 and 6.9 minutes respectively).
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Recommendations

- Implement user engagement reminders and incentives (e.g., first ride discount or promo codes) to convert registered users into active riders.
 - Increase driver availability during peak hours to reduce acceptance time and increase revenue (e.g., driver bonuses).
 - Encourage users to leave feedback after rides to improve rating coverage.
 - Make age a required field during registration to improve user analytics.
 - Improve handling of negative reviews by collecting structured feedback when ratings are below 3 stars.
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In this project, I analyzed the data using SQL and created visualizations using Tableau Public.